BALL STATE UNIVERSITY

Practical Assignment IV

Introduction to Statistical Methods

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1 Pre-assignment Reminders

- Prior to attempting this homework, I strongly recommend that you have completed all other work through module 12, including:
  - Having watched all lectures thoroughly and taken notes (optionally with the guided notes)
  - Use the posted SPSS tutorials, near the top of Canvas; there are also additional SPSS tutorials listed under "Student Support & Additional Resources"
  - Reading all assigned sections of the book, and completing practice examples as helpful and useful
  - Completing relevant lecture check-ins, and using answer key for review
  - Completing all quizzes, and reviewing correct answers at the end of the week
- This practical assignment will be graded for accuracy, please give yourself enough time to complete your best work
  - You may use your notes, book, SPSS software, provided data, and lectures to aid you in completing this
  - You may also use other resources, but always double-check your work and ensure that results "make sense"
  - Where possible, include more detail to ensure you fully explain your rationale for a question
  - You should include ALL relevant SPSS output to support your work, please make clear where in the output you get your answers from
- This practical assignment is cumulative to the modules and practical assignment(s) prior, please review your previous work and notes to help you accomplish this work. Questions asked in this assignment may assess your knowledge in prior areas
- All syllabus and university policies on academic integrity, plagiarism, and other forms of misconduct apply to this assignment. Please review them if you are unfamiliar

> **Important**
>
> Remember, it is critically important to me that you can explain your work and how it was accomplished (even if you are confused)!

2 Context

The following information will be useful for answer some questions on the homework, please ensure you read and understand the following context

You are a principal in a small, rural school district and are trying to assess contributors to high school student scores on a recent, state-wide assessment. The assessment is very important, as schools showing consistent, high performance on the test will be allocated more funds in an upcoming budget. You notice large variability in student performance, so you need to diagnose what factors may be related to issues and success in doing well on the statewide assessment.

You have a data set with several variables that you can use to start a preliminary analysis:

- Student's sex (Male or Female)
- Student is in advanced or regular classes (Advanced or Regular)
- Student is intending to attend college (Yes or No)
- Student's year in high school (First, Second, Third, Fourth)
- Student's score on statewide exam (0 to 500; continuous)
- Student's household annual income (in dollars)
- Student's cumulative GPA (from 0.00 to 4.00; continuous)

This data will be presented in assessment_data.sav on the Canvas page.

3 Instructions

> **Important**
>
> Please - provide as much detail as you can to adequately address each question; I'd prefer more information rather than less. Remember to follow all expectations for creating original work, as described in the syllabus.

Answer all of the following questions in full, using appropriate vocabulary from past modules:

3.1 Use Your Resources! (4pts)

It is important to me that you understand and demonstrate that there are many useful tools, methods, and guides to helping complete good analyses, while learning at the same time! Earn some easy points here by selecting several additional resources (outside of me, the professor) to aid you in this assignment. While I'd like to believe I can be helpful to you, I know there are many great ways to learn.

You may do this section last after you've worked through the rest.

1. Identify one resource you used to complete this assignment, under the "Additional SPSS Written Guides" module in Canvas (near the top of the Canvas page!; 1pt)

2. Identify one resource you used to complete this assignment, under the "SPSS Tutorials & Resources" Header in the "Student Support & Additional Resources" module module in Canvas (near the top of the Canvas page!; 1pt)

3. Find one outside resource not linked on Canvas, to help you complete this assignment - any website, tool, or resource is fair game. Provide me a link to this resource (1pt)
4. Reflect on whether each of these resources helped you more, less, or the same as my usual video demonstrations that accompany these practical assignments. It's okay to be honest! (1pt)

3.2 Knowledge Checks (6 pts)

*For the following, please make sure you explain **why** each test is appropriate to the hypothesis you've presented.*

1. Using the available variables in the dataset, create an alternative-null hypothesis pair that will require use of a chi-squared analysis (2pts)
2. Using the available variables in the dataset, create an alternative-null hypothesis pair that will require use of a correlation or linear regression (your choice of which one; 2pts)
3. Using the available variables in the dataset, create an alternative-null hypothesis pair that will require use of a t-test (your choice of which one; 2pts)

3.3 SPSS Applied Work (10 pts)

For each of the following questions, please provide any and all relevant SPSS output (or syntax when indicated). Follow the directions IN ORDER

1. Show the SPSS syntax you'd use to make a new variable that splits students into two group, one who passed the statewide assessment (score of 350 or above) and those who failed (score of 349 or lower). Run this syntax prior to next steps (1pt)
2. Show the SPSS syntax you'd use use to filter out all first-year students. Run this syntax prior to next steps (1pt)
3. Show the descriptive statistics for ALL variables in the dataset. For continuous variables, show the mean, median, and standard deviation, as well as a frequency histogram. For categorical variables, show a frequency table and a bar graph. (2pts)
4. Show the syntax and results for the chi-square analysis you planned in Knowledge Checks (2pts)
5. Show the syntax and results for the correlation or linear regression you planned in Knowledge Checks (2pts)
6. Show the syntax and results for the t-test you planned in Knowledge Checks (2pts)